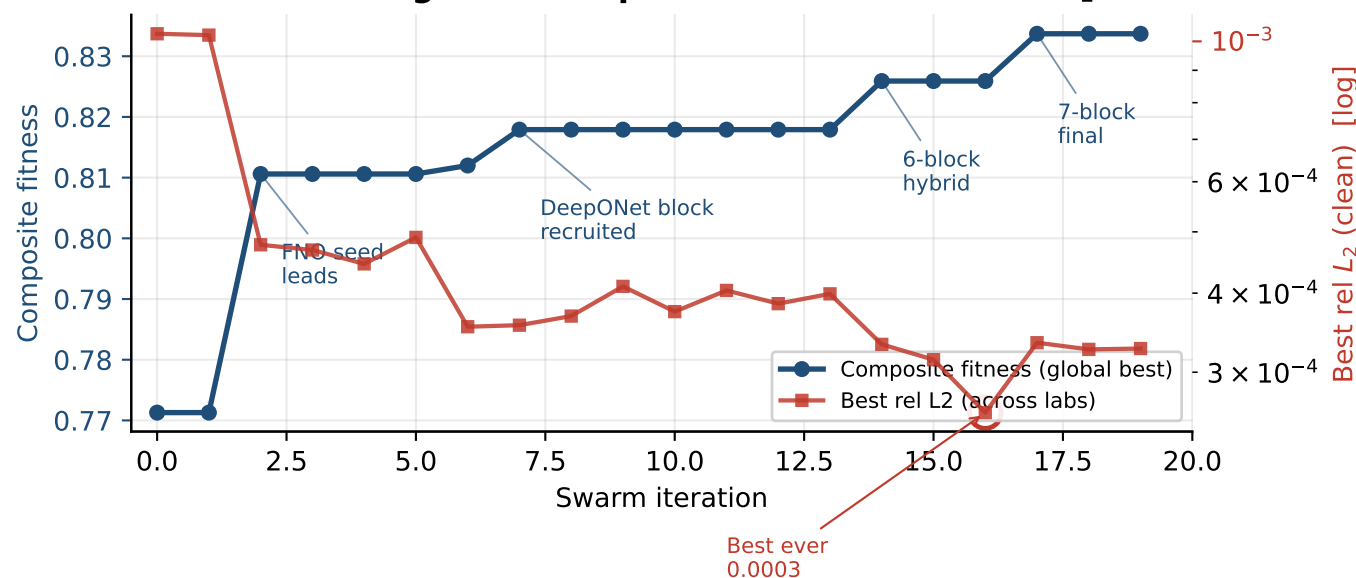


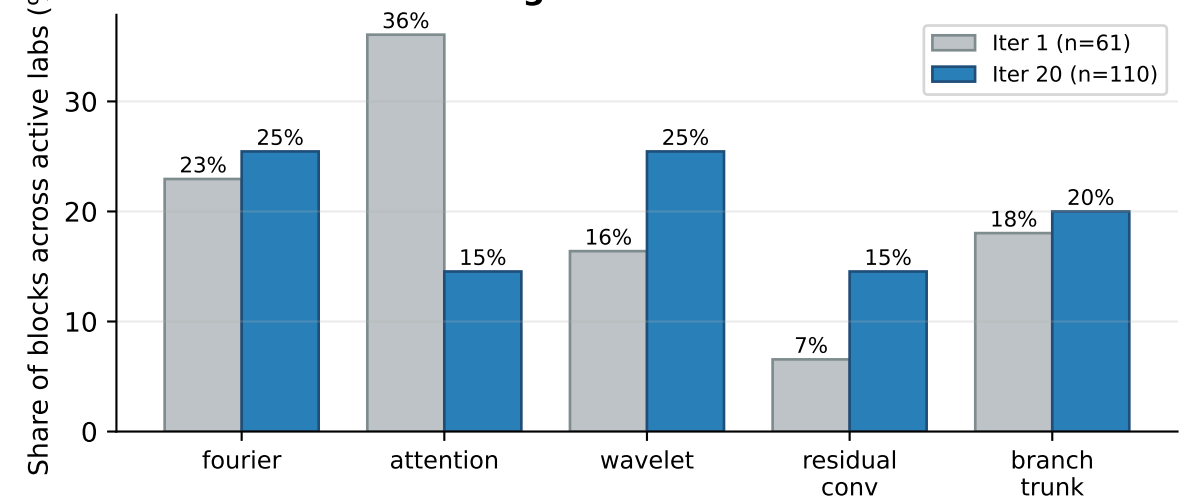
Agentic Neural Operator Discovery: 2D Navier-Stokes (paper-grade run, seed 42)

16 virtual labs × 20 iterations × 50 epochs × 512 train samples • PSO swarm with citation-based peer review

A. Convergence: composite fitness + best rel L_2

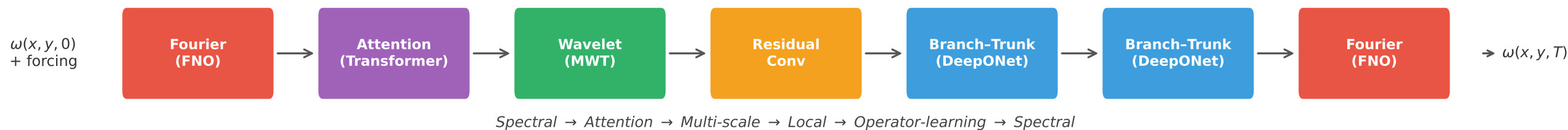


B. Block usage shift: swarm consensus

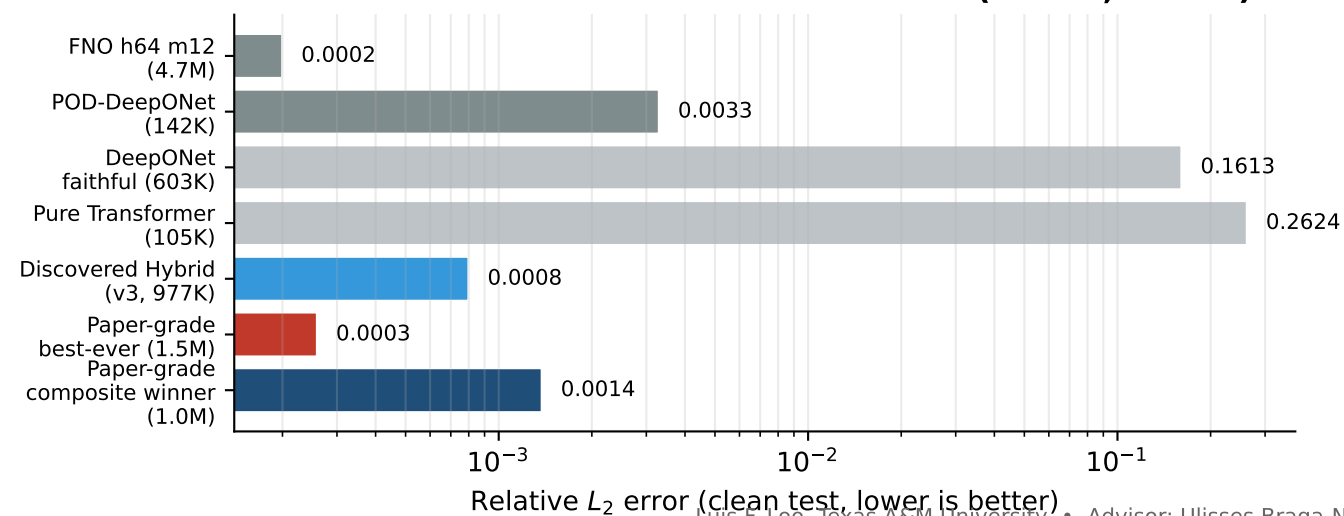


C. Discovered architecture (paper-grade, NS-2D, seed 42)

Discovered hybrid (lab 13, fitness 0.8337, hidden=64, modes=10, GELU, ~ 1.0M params)



D. Where the swarm sits vs baselines (NS-2D, res 32)



E. Take-home (advisor-safe framing)

Honest claim for ICERM

- Agentic discovery converges on a regime-aware hybrid that recombines four operator families (Fourier + Attention + Wavelet + Branch-Trunk) with no human prior on architecture.
- Best-ever rel $L_2 = 2.6e-4$ with ~1.5M params, vs 4.7M for the FNO baseline (~3× smaller).
- Block usage shifts measurably between iter 1 and iter 20 $\rightarrow$ swarm is not random search.
- Cross-PDE comparison (Darcy 2-D) is the natural next run; pending advisor approval.

We do NOT claim to beat DeepONet – POD-DeepONet (142K) still wins on parameter efficiency, and DeepONet's regime is Darcy-style operators.